

V3 Pipeline — Egitim Raporu

Tarih: 2026-03-22 21:49

PIPELINE OZELLIKLERI:

Feature Engineering: V3 (Label-encoded predictions + ayri OHE + manuel k-mer)
Feature sayisi: 976
Toplam ornek: 4287 (CV: 3429, Hold-Out: 858)
Model turleri: LightGBM, XGBoost
Hyperparameter: Grid Search (12 combo/model, 3-fold CV)
Degerlendirme: 5-Fold Stratified CV + %20 Hold-Out Test

5-FOLD CV SONUCLARI (egitim seti %80):

lightgbm : F1 = 0.9765 +/- 0.0027
xgboost : F1 = 0.9742 +/- 0.0018

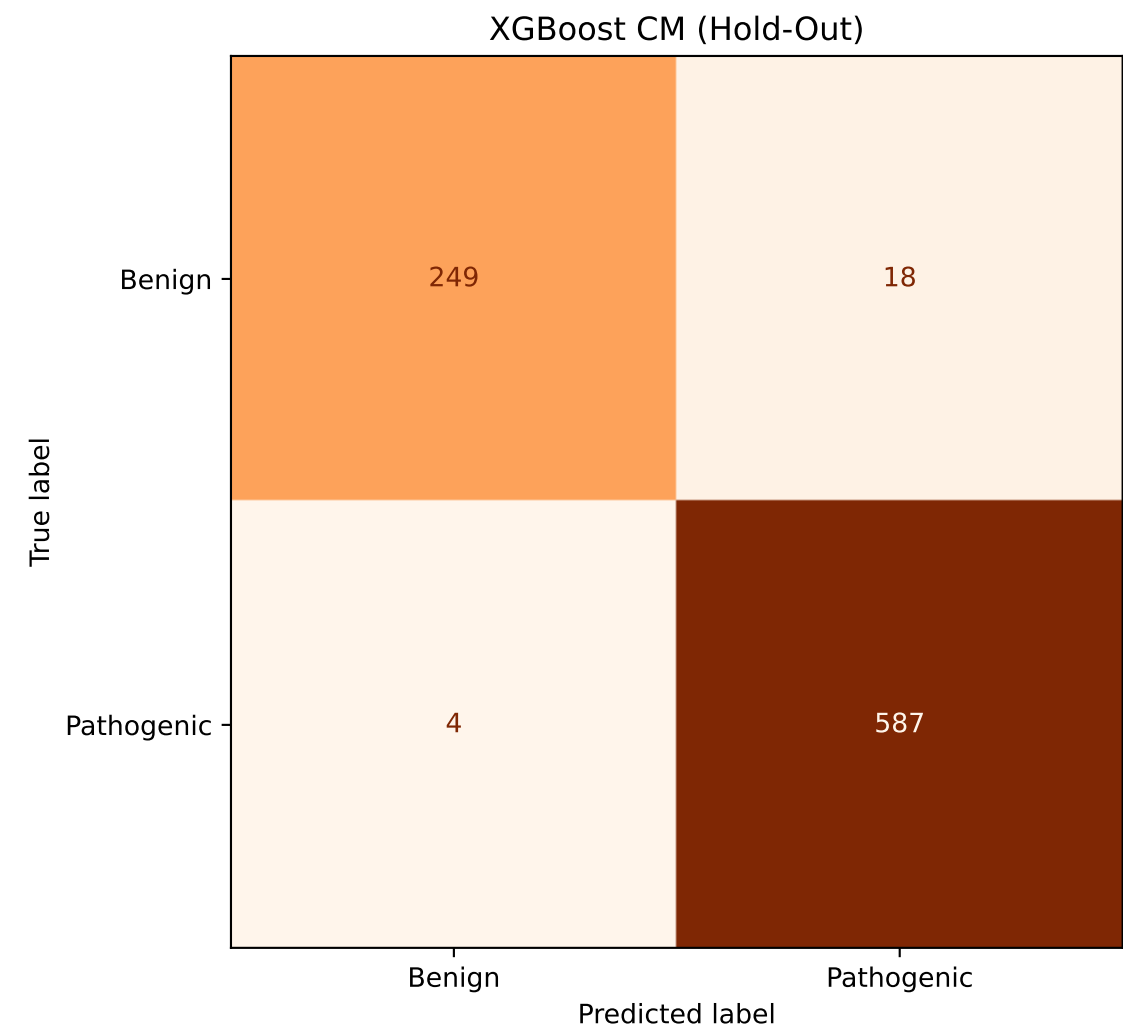
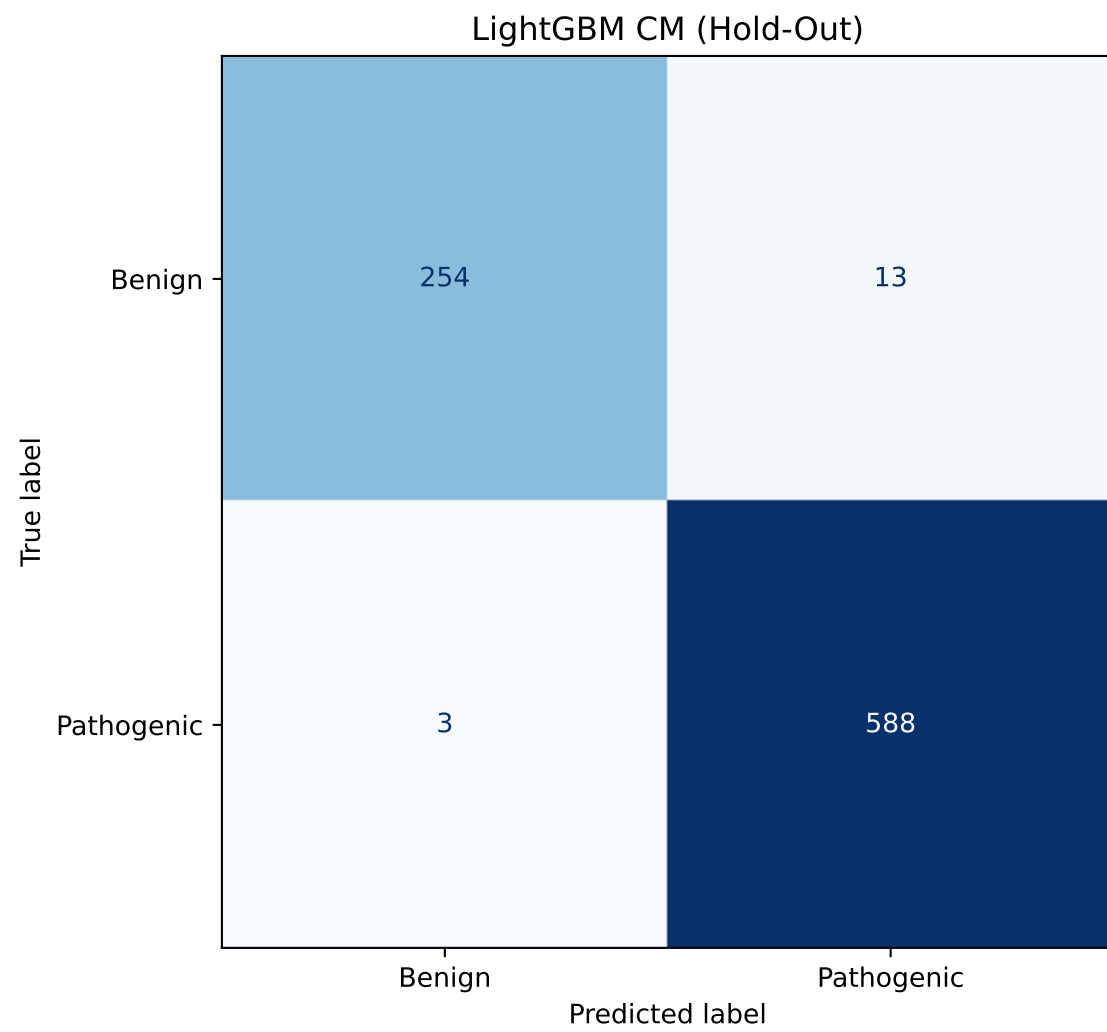
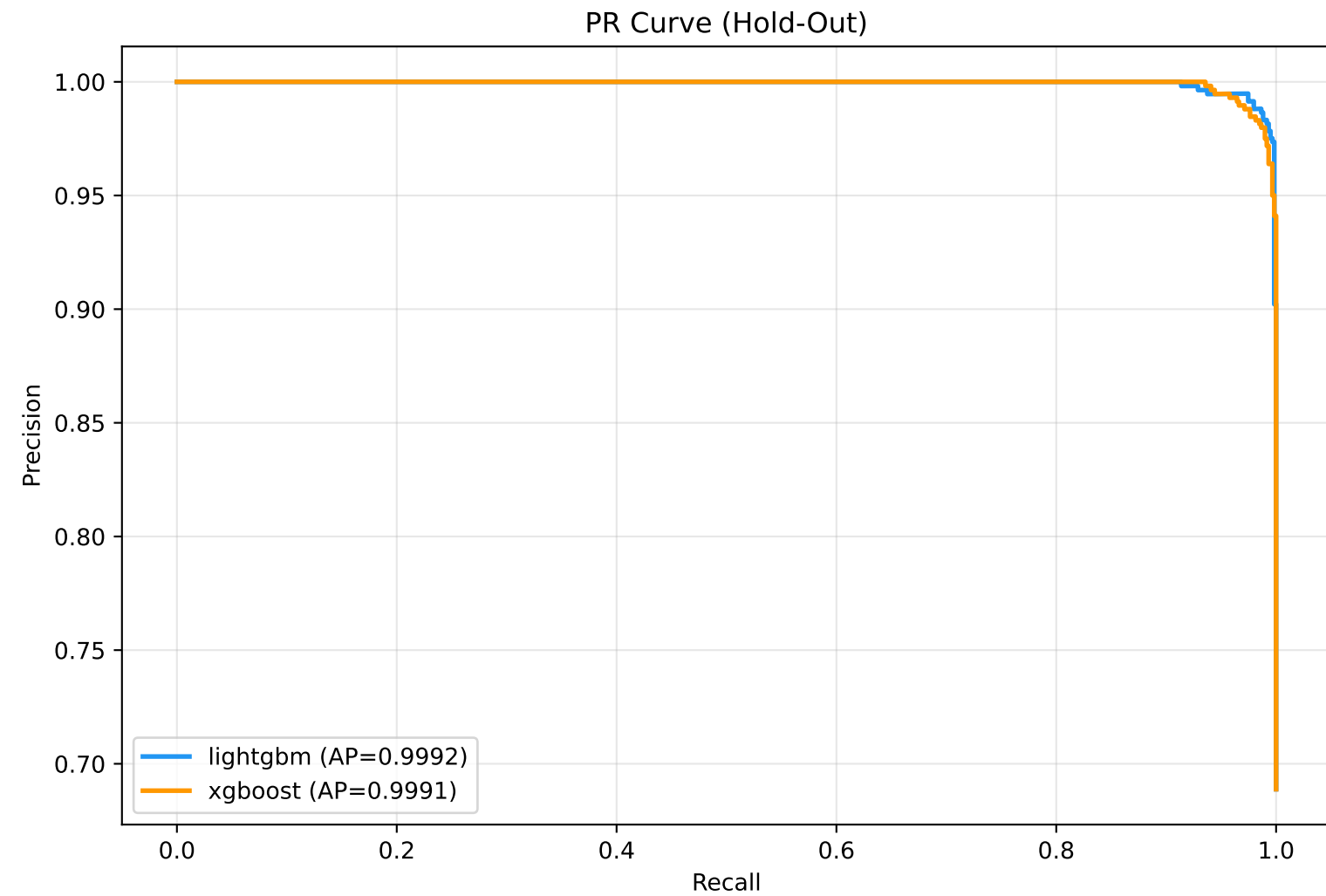
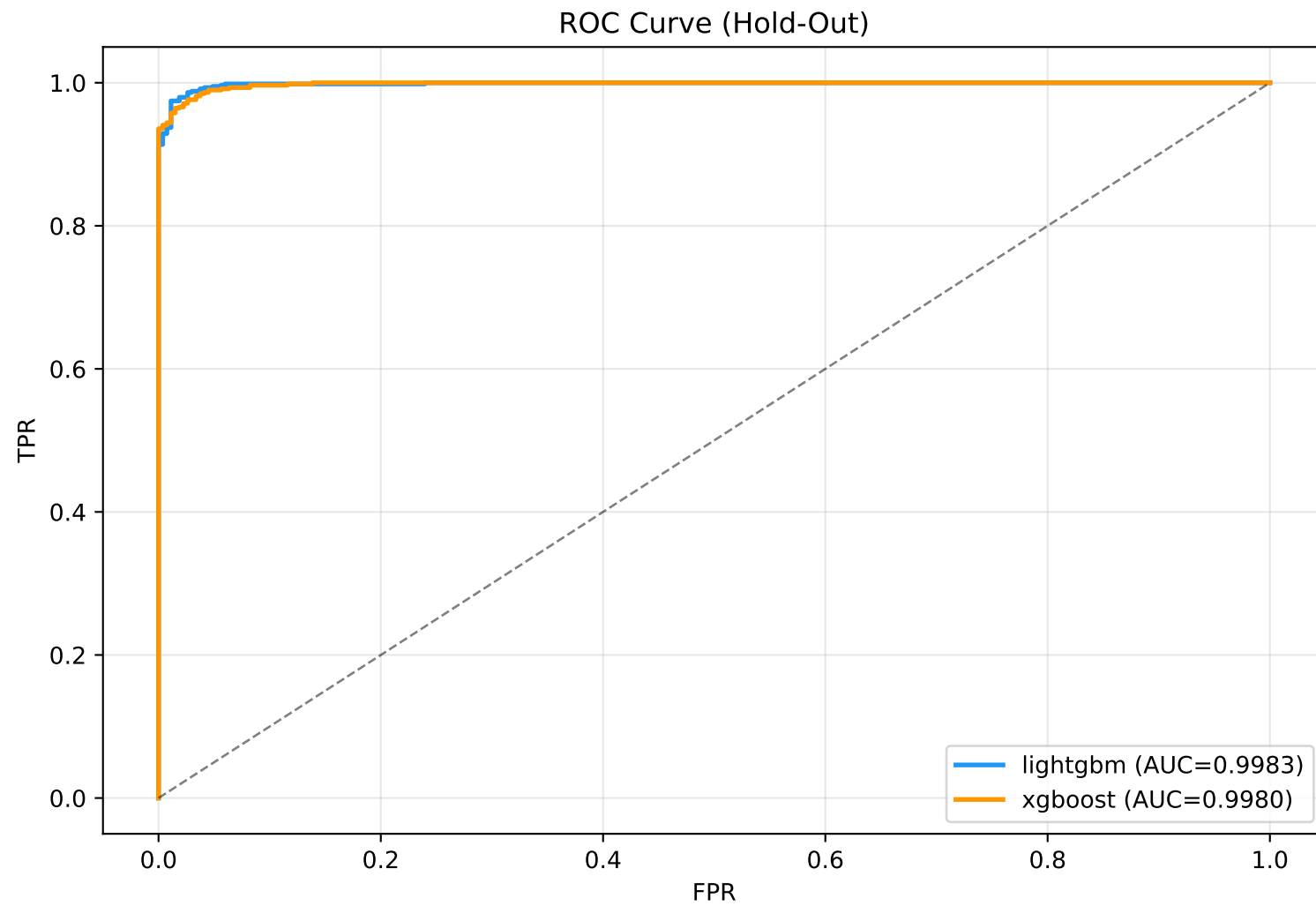
HOLD-OUT TEST SONUCLARI (hic gorulmemis %20):

lightgbm : F1=0.9866 AUC=0.9983 (CV F1=0.9765)
xgboost : F1=0.9816 AUC=0.9980 (CV F1=0.9742)

KARSILASTIRMA (full_fe vs v3 hold-out):

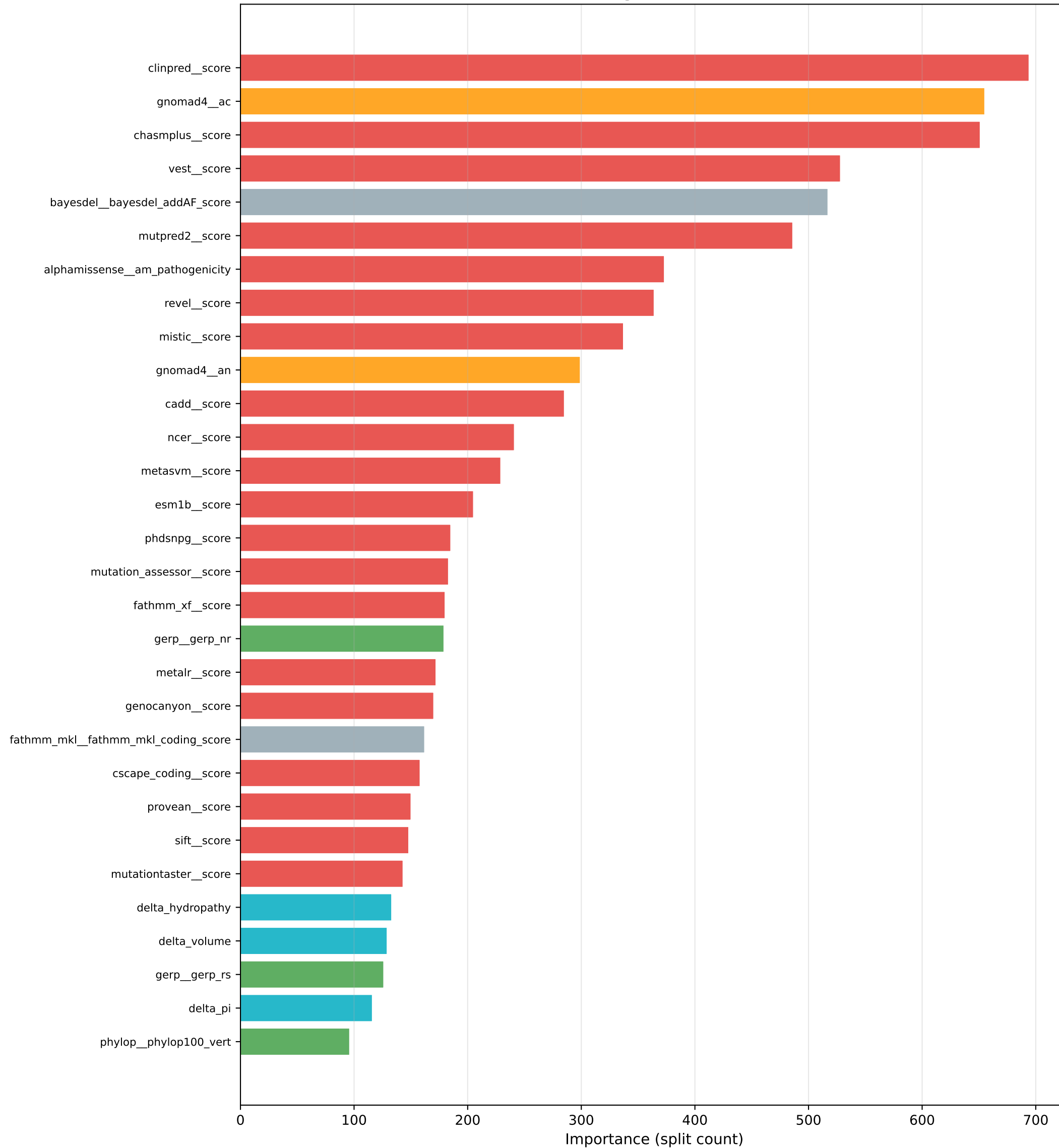
lightgbm: full_fe F1=0.6967 -> v3 Hold-Out F1=0.9866 (+0.2899)
xgboost: full_fe F1=0.6879 -> v3 Hold-Out F1=0.9816 (+0.2937)

Hold-Out Test Sonuclari (%20 — Hic Gorulmemis Veri)

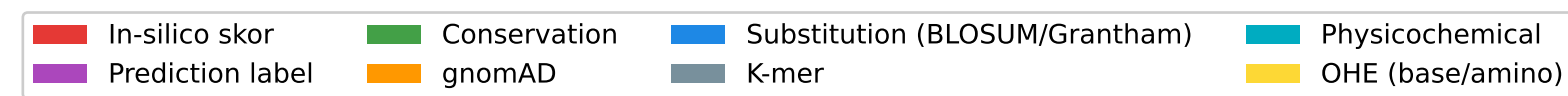
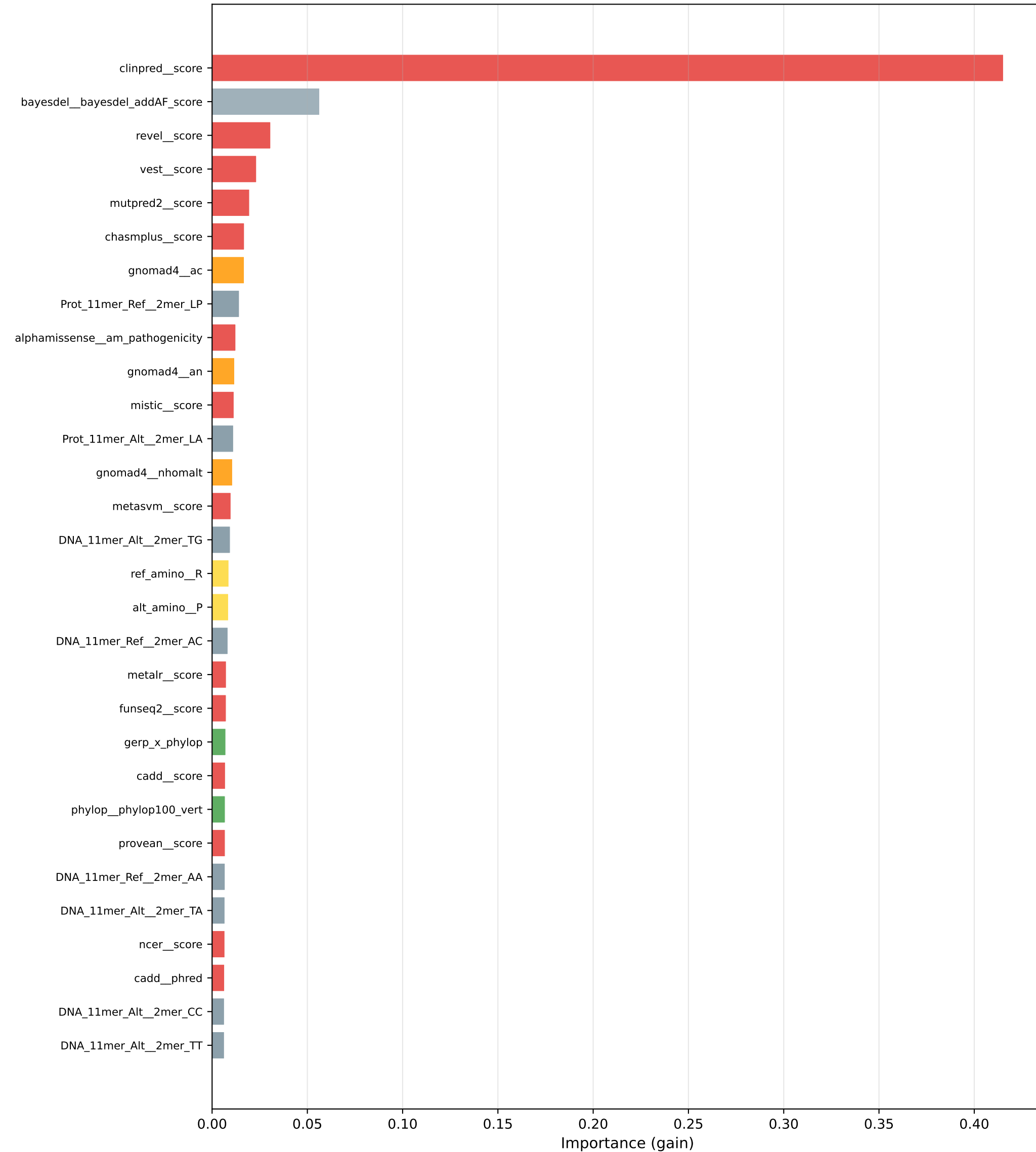


Feature Importance — Top 30 (En İyi CV Fold Modeli)

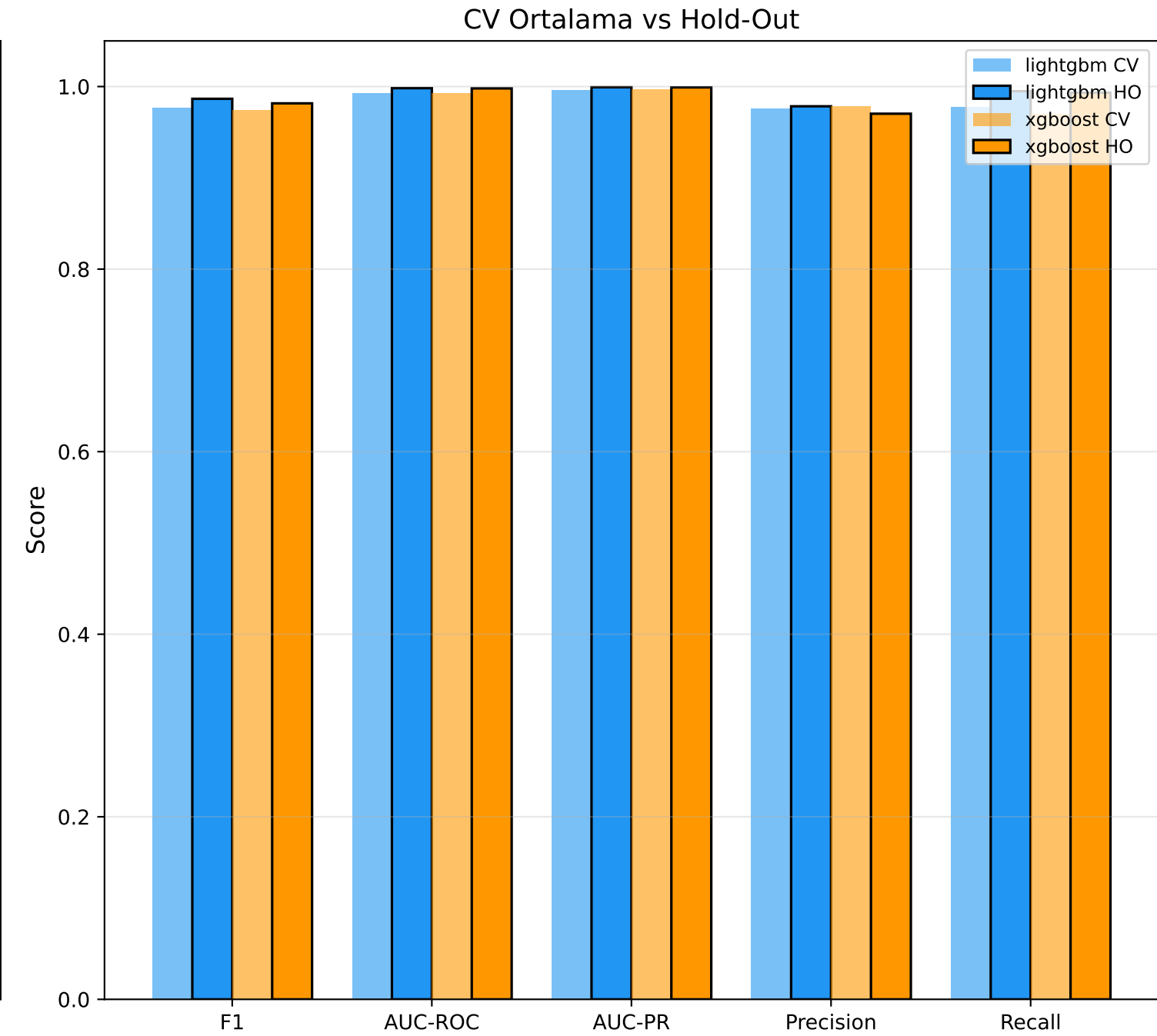
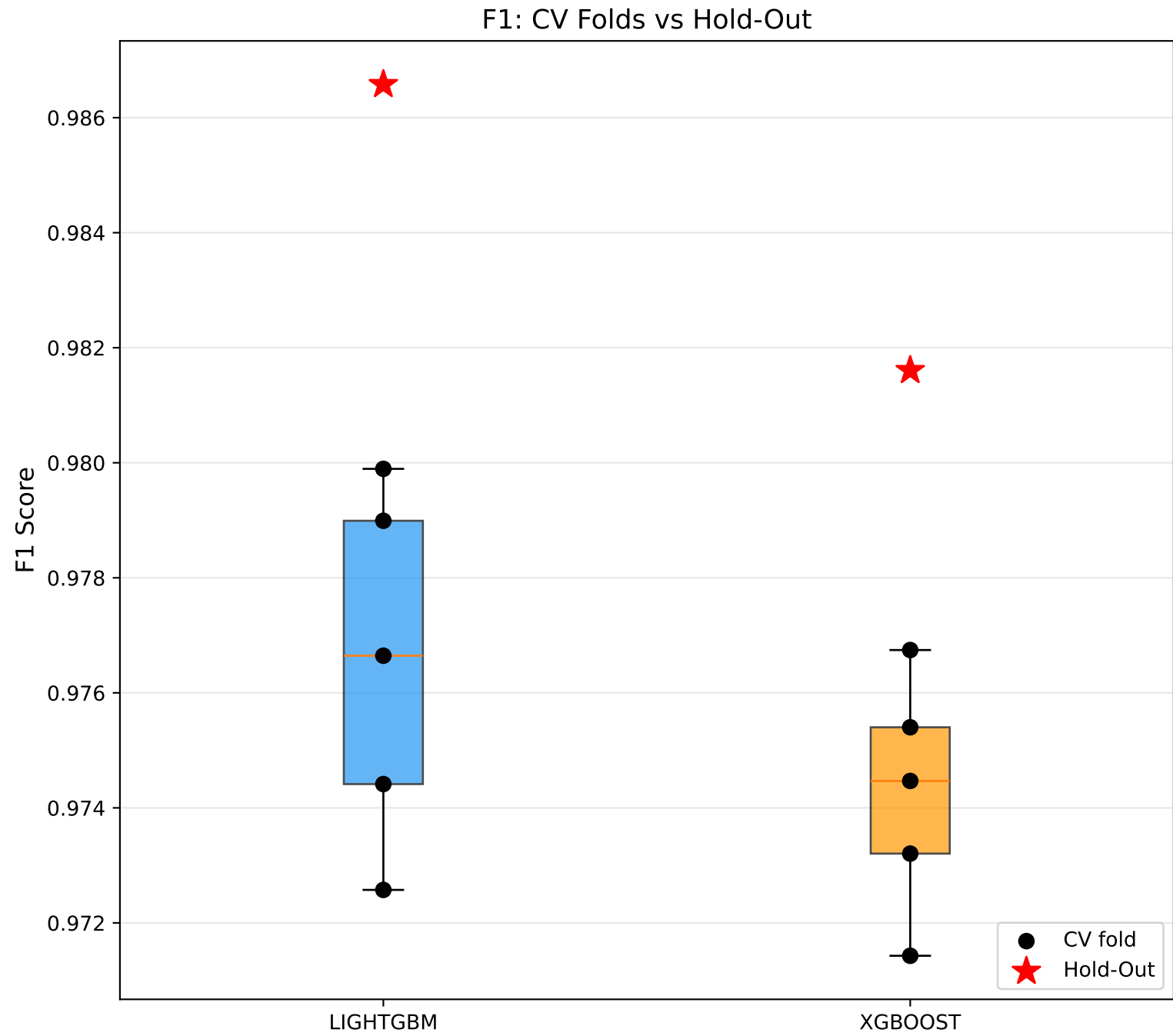
LightGBM



XGBoost



CV vs Hold-Out Karsilastirma



Detayli Sonuclar: 5-Fold CV + Hold-Out Test

Fold	Model	Split	F1	AUC-ROC	AUC-PR	Precision	Recall	MCC
1	lightgbm	CV	0.9790	0.9924	0.9966	0.9708	0.9873	0.9317
2	lightgbm	CV	0.9744	0.9905	0.9948	0.9807	0.9682	0.9193
3	lightgbm	CV	0.9799	0.9921	0.9948	0.9789	0.9809	0.9354
4	lightgbm	CV	0.9766	0.9944	0.9976	0.9787	0.9746	0.9255
5	lightgbm	CV	0.9726	0.9934	0.9970	0.9685	0.9767	0.9111
1	xgboost	CV	0.9767	0.9938	0.9973	0.9747	0.9788	0.9251
2	xgboost	CV	0.9732	0.9905	0.9948	0.9848	0.9619	0.9169
3	xgboost	CV	0.9745	0.9920	0.9959	0.9786	0.9703	0.9190
4	xgboost	CV	0.9754	0.9950	0.9978	0.9849	0.9661	0.9232
5	xgboost	CV	0.9714	0.9932	0.9969	0.9704	0.9725	0.9079
ORT	lightgbm	CV	0.9765	0.9926	0.9961	0.9755	0.9775	0.9246
ORT	xgboost	CV	0.9742	0.9929	0.9965	0.9787	0.9699	0.9184
-	lightgbm	HOLD-OUT	0.9866	0.9983	0.9992	0.9784	0.9949	0.9564
-	xgboost	HOLD-OUT	0.9816	0.9980	0.9991	0.9702	0.9932	0.9400